Partial Solutions To Exercises In Group Theory

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October 2, 2026

Intro

Solutions the HW1-10 from 24B in group theory. Each section represent a HW.

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. . . . . (1) . . . . .

1. (a)

   (b) $G \times H$ is Abelian iff G, H are both Abelian

   Proof. Indeed:

   $\Longrightarrow$ If $G \times H$ is Abelian, so π_1, π_2 are group homomorphisms (which preserve group properties like being Abelian) therefore its images G, H are also Abelian.

   $\Longleftarrow$ If G, H are Abelian, and let $(g_x, h_x) = x, (g_y, h_y) = y \in G \times H$, then:

   $$\begin{aligned} x \cdot y &= (g_x, h_x) \cdot (g_y, h_y) = (g_x \cdot g_y, h_x \cdot h_y) \\ &= (g_y \cdot g_x, h_y \cdot h_x) \quad \xleftarrow[G, H \text{ are Abelian}]{} \\ &= (g_y, h_y) \cdot (g_x, h_x) \\ &= y \cdot x \end{aligned}$$

   As intended.

   ■

2. Assume $\forall g \in G\colon g^2 = e$. We'll prove G is Abelian.

   Proof. Let $x, y \in G$. Clearly $x = x^{-1} \wedge y = y^{-1}$ (since $y^2 = x^2 = 1$). Then $(xy)^2 = e$ therefore $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$ as intended. ■

3. Consider the group $G = \mathrm{L}_2(\mathbb{R})$ (the linear space of dimension two with addition) and the groups $H_1 = \mathrm{span}((1,0)), H_2 = \mathrm{span}((0,1))$. Clearly H_1, H_2 are subgroups (since they're sub-vector-spaces), though $H_1 \cup H_2$ isn't not a group since $(1,0) + (0,1) \notin H_1 \cup H_2$ (addition not closed).

4. Consider $\mathrm{SL}_2(\mathbb{Z})$. We'll show its a subgroup of $\mathrm{GL}_2(\mathbb{R})$.

   Proof. Clearly $\mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{L}_2(\mathbb{R})$. Though since if $\det A = 1$ then A is invertible, we know that $\mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{GL}_2(\mathbb{R})$. It remains to show that $\mathrm{SL}_2(\mathbb{Z})$ is a group. It's clear since if $A, B \in \mathrm{SL}_2(\mathbb{Z})$ then $\det AB = \det A \det B = 1$ and the coefficients of AB must lie in $\mathbb{Z}$ (matmul definition) therefore it's closed another multiplication. Similarly $\det(A^{-1}) = (\det A)^{-1} = 1$ and $A^{-1} = \frac{\mathrm{adj}\, A}{\det A} = \mathrm{adj}\, A$, and clearly the determinant of the minors of A are integers (an integer polynom that gets an integer value yields and integer result) therefore $A^{-1} = \mathrm{adj}\, A \in \mathrm{SL}_2(\mathbb{Z})$ too.

   We already know $\mathrm{GL}_2(\mathbb{R})$ multiplication satisfies the conditions for group multiplication, so $\mathrm{SL}_2(\mathbb{Z})$ must form a subgroup of $\mathrm{GL}_2(\mathbb{R})$. ■

5.

6.

(2)

1. We'll prove that $g^n = 1 \iff \text{ord}(g) \mid n$.

   Proof. If $\text{ord}(g) \mid n$ then $k \cdot \text{ord}(g) = n$ for some $k \in \mathbb{N}_0$. Almost by definition $g^{\text{ord}(g)} = e$ and now:

   $$g^n = g^{\text{ord}(g)\cdot k} = \left(g^{\text{ord}\, g}\right)^k = 1^k = k$$

   As intended.

   The other side will be proven via contrapositive. AFSOC $\text{ord}(g) \nmid n \wedge g^n = 1$. Then consider the division with (minimal) remainder $n = k\,\text{ord}(g) + r$ $(0 < r < \text{ord}(g))$. Then we get that:

   $$1 = g^n = g^{k\,\text{ord}(g)+r} = g^r \cdot \left(g^{\text{ord}(g)}\right)^k = g^r$$

   So we found r s.t. $g^r = 1$, and $r \neq 0$. Therefore $\text{ord}(g) \leq r$, yet $r < \text{ord}(g)$, which yields a contradiction. ■

2. Consider $G = \langle g \rangle$ a cyclic group generated by g with order $|G| = n$. Let $d \mid n$. Then:

   - $H = \{x \in G \colon x^d = 1\}$ is a subgroup with order d.

     Proof. Clearly H is a subgroup (if $x^d = 1 \wedge y^d = 1$ then $(xy)^d = x^d y^d = 1$ and $(x^{-1})^d = (x^d)^{-1} = 1^{-1} = 1$). We know that $\exists k \in \mathbb{N}_0 \colon dk = n$. Now consider $p \coloneqq g^k$. Clearly $p^d = p^{kd} = p^n = 1$. Now consider the subgroup $H = \langle p \rangle$. It's a cyclic subgroup with order of d, since $\text{ord}(p) = d$. We'll show that $d \in H$: H must be cyclic (subgroup of a cyclic group must be cyclic) therefore it's generated by some element p'. Note that $p' = p$ otherwise $p'^z = p$ for $z > 1$ yet it means that $p^{d-z} = 1$ although $\langle p \rangle$ has an order of d. Hence indeed $H = \langle p \rangle$, which has the order d. ■

   - We'll show it's the only subgroup with order d.

     Proof. Consider the existence of a (cyclic) group H' with order d. Then it's generated by some p' s.t. the minimum argument ℓ s.t. $p'^\ell = 1$ is $\ell = d$. This strictly implies $p = g^k$ (with the notation $dk = n$ above) as $d \mid n$ (it's given but also known more generally by Lagrange). ■

3.

4. We'll prove $(\mathbb{Q}, +)$ is not finitely generated.

   Proof. If $(\mathbb{Q}, +)$ was finitely generated by some $g_1 \dots g_n$ where $g_i = \frac{p_i}{q_i}$ $(\gcd(p_i, q_i) = 1)$ then the number $g = \frac{1}{1+\sum_{i=1}^n q_i} \in \mathbb{Q}$ could have not been reached since the denominator of g must be a multiplication of the q_i-s (which of course it isn't). We found $g \in \mathbb{Q}$ but not in the group generated by $g_1 \dots g_n$, and via deductio ad absurdum we finished. ■

5. We'll prove the double-costs are a partition of G.

   Proof. content ■

(3)

1. We need to calculate $3^{199} \bmod 101$. Notice that $3^{101} = 3 \mod 101$ therefore in $\mathbb{Z}/101\mathbb{Z}$ we know that $3^{202} \equiv_{101} 3^2 \equiv 9$. We aim to find x s.t. $x \cdot 3^3 \equiv= 101k + 9$. Going through the options:

   - For $k = 1$ we get $110 \nmid 3^3$
   - For $k = 2$ we get $221 \nmid 3^3$
   - ...
   - For $k = 9$ we get $918 \mid 3^3$

   Therefore $x \cdot 3^3 \equiv 918 \implies x \equiv \frac{918}{3^3} = 34$.

2. Let G be a group, and H, K subgroups. We'll prove that $[H : H \cap K] \leq [G : K]$.

Proof. Consider the function $\varphi\colon H/(H\cap K)\to G/K$ defined as $\varphi(kH)=kG$ (where $k\in H\cap K$). It's clearly independent of choice of representative k since if $(H\cap K)k_1=(H\cap K)k_2$ then $Hk_1=Hk_2$ (since $k_1k_2^{-1}\in H\cap K\subset H$ in this case). Moreover, φ is injective since if $Hk_1=Hk_2$ then $(H\cap K)k_1=(H\cap K)k_2$ ■

3. Let G be a group generated by $S\subset G$. Let $\varphi,\psi\colon G\to H$ be homomorphisms s.t. $\forall s\in S\colon \varphi(s)=\psi(s)$. Then $\varphi=\psi$.

   Proof. Let $g\in G$. By definition, exists $s_1\dots s_n\in S$ s.t. $s_1+\dots+s_n=g$. Therefore:

   $$\forall g\in G\colon \varphi(g)=\varphi\left(\sum_{i=1}^{n}s_i\right)=\sum_{i=1}^{n}\varphi(s_i)=\sum_{i=0}^{n}\psi(s_i)=\psi\left(\sum_{i=0}^{n}s_i\right)=\psi(g)$$

   ■

4. We'll find all of the homomorphisms $f\colon G\to H$ in the following cases:
   (a) Let $G=H=\mathbb{Z}$. Claim: $f(z)=az$ for some $a\in\mathbb{Z}$ (f is the quotient map that maps $\mathbb{Z}$ into the normal subgroup $a\mathbb{Z}$). Clearly f is an homomorphism, we'll proof it's the only homomorphism. Indeed, considering some $f\colon\mathbb{Z}\to\mathbb{Z}$ homomorphism, denote $f(1)=a$. Then for each $z\in\mathbb{Z}$, because $\mathbb{Z}$ is finitely generated by 1 ($z=1+1+\dots+1=:1\cdot z$) we know that $f(1\cdot z)=z\cdot f(1)$ by definition. Now $f(z)=az$ as intended.

   [btw, f is a ring isomorphism too]

   (b) Let $G=\mathbb{Z},H=\mathbb{Z}/n\mathbb{Z}$. Claim: $f(z)=[anz]$ for $a\in\mathbb{Z}$ are the only homomorphisms $\mathbb{Z}\to\mathbb{Z}/n\mathbb{Z}$. It's clearly a homomorphisms. And the proof the existence is similar to the one in (A)

   (c)

   (d)

5. Proof/Disproof the following claims:
   - Claim: $(\mathbb{Z},+)\cong(\mathbb{Q},+)$.

     Disproof. Consider the set of generators of $\mathbb{Z}$, which is $S=\{1_{\mathbb{Z}}\}$. Assume for the sake of contradiction that there exists and isomorphism $f\colon\mathbb{Z}\to\mathbb{Q}$. Therefore $f(S)$ is the set of generators of $\mathbb{Q}$. Identities are send to identities by isomorphisms and thereafter $f(S)=\{1_{\mathbb{Q}}\}$. But $\frac{1}{2}\in\mathbb{Q}$ is clearly not generated by $1_{\mathbb{Q}}$, and ad absurdum the claim is false. ■

   - Claim: $(\mathbb{Q},+)\cong(\mathbb{Q}^\times,\cdot)$.

     Disproof. In $\mathbb{Q}^\times$ there exists a root os unity -1 (since $-1\cdot-1=1$). Therefore the isomorphism $f\colon\mathbb{Q}\to\mathbb{Q}^\times$ preserve the root of unity $f(-1)\cdot f(-1)=1_{\mathbb{Q}}=0$. But in $\mathbb{Q}$ there are not root of unity (if $a+a=0$ then $2a=0$ and $a=0$) therefore $f(-1)=0$, meanwhile $f(1)=f(1_{\mathbb{Q}^\times})=1_{\mathbb{Q}}=0$ therefore f is not injective, which is a contradiction. ■

6. We'll prove that up to an isomorphism, there's only one group with the order of p (p is prime).

   Proof. It's sufficient to prove that every group with a prime order must be cyclic, since all of the finite cyclic groups isomorphic to $\mathbb{Z}/p\mathbb{Z}$.

   Let G be a finite group with the prime order of p. Consider the cyclic subgroup $H\leq G$ generated by $a\neq 1_G$ (G is nontrivial since $p\neq 1$ therefore such a must exists). By Lagrange $\operatorname{ord}G=\operatorname{ord}H\cdot[G:H]$. But $\operatorname{ord}G=p$ is prime, therefore either $\operatorname{ord}H=1$ or $\operatorname{ord}H=p$. But $a,1_G\in H$ therefore $\operatorname{ord}H\neq 1$. Finally $\operatorname{ord}H=\operatorname{ord}G=p$, hence $H=G$ and G is cyclically generated by a. ■

(4)

(5)

1. (a) We'll prove $\forall\mathbb{N}\ni n\geq 1\colon(\mathbb{Z}\times\mathbb{Z})/\langle(n,n)\rangle\cong\mathbb{Z}\times(\mathbb{Z}/n\mathbb{Z})$.

Proof. Let $n \in \mathbb{N}_+$. The questions is obviously well-defined since $\mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z}$ are Abelian.

Consider the map $\varphi\colon (\mathbb{Z} \times \mathbb{Z}) \to (\mathbb{Z} \times (\mathbb{Z}/n\mathbb{Z}))$ defined by:

$$\varphi(x, y) \coloneqq (x, y \bmod n)$$

It's clearly an homomorphism, onto (it's the identity under the trivial embedding of $\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ into $\mathbb{Z} \times \mathbb{Z}$), and well-defined ($y \bmod n \in \mathbb{Z}/n\mathbb{Z}$). Moreover:

$$\ker \varphi = \left\{(x, y \bmod n) = (0, 0) \mid (x, y) \in \mathbb{Z}^2\right\} = \{(0, nk) \mid k \in \mathbb{Z}\} = \langle (0, n) \rangle \cong \langle (n, n) \rangle$$

So by the first isomorphism theorem, we finished the proof. ∎

2. Consider G to be a finite group and $H \neq N \leq G$ s.t. $\operatorname{ord} G = \operatorname{ord} H = p$ prime, and $N \lhd G$.
   (a) We'll first prove that $G = HN$. By Lagrange $|G| = p \cdot [G : H] = p \cdot [G : N]$.
3. Consider $\operatorname{Aut} G$.
   (a) It's indeed a group:

   Proof. Let $\varphi, \psi\colon G \to G$ isomorphism. Then $\varphi \circ \psi$ must be an isomorphism too since:

   $$(\varphi \circ \psi)(xy) = \varphi(\psi(x)\psi(y)) = (\varphi \circ \psi)(x) \cdot (\varphi \circ \psi)(y)$$

   And a composition of injective functions results in an injective function. Moreover, since φ is an injective there exists φ^{-1} well defined, which similarly must be an isomorphism too. ∎

   (b) We'll prove $\operatorname{Inn} G \lhd \operatorname{Aut} G$.

   Proof. Consider an inner automorphism $\varphi_a(x) \coloneqq a^{-1}xa$ and an automorphism $\psi\colon G \to G$. Denote $u \coloneqq \psi^{-1}(a)$. Then:

   $$(\psi^{-1} \circ \varphi_a \circ \psi)(x) = \psi^{-1}(\varphi_a(\psi(x))) = \psi^{-1}(a^{-1}\psi(x)a) = \psi^{-1}(a^{-1}) \cdot \psi^{-1}(\psi(x)) \cdot \psi^{-1}(a) = u^{-1} \cdot x \cdot u = \varphi_u$$

   Since $\psi^{-1}(a^{-1}) = \left(\psi^{-1}(a)\right)^{-1} = u^{-1}$ as ψ is an inner automorphism and in particular homomorphism. We just proved that $\psi^{-1} \circ \varphi_a \circ \psi$ is an inner automorphism (in particular φ_u), hence $(\operatorname{Inn} G)^{\psi} \subset \operatorname{Inn} G$. It remain the prove a reverse inclusion. Indeed, we already notice that $\varphi_a \mapsto \varphi_{\psi^{-1}(a)}$, and therefore $\varphi_{\psi(a)} \mapsto \varphi_a$ (or in other words, $\varphi_{\psi(a)} \in \operatorname{Inn} G$ implies the existence of $\varphi_{\psi^{-1}(\psi(a))} = \psi(a)$ in $(\operatorname{Inn} G)^{\psi}$). As intended. ∎

   (c) We'll compute $\operatorname{Aut}(C_p)$ and $\operatorname{Inn}(C_p)$ for $p \in \mathbb{N}$ prime.

   We've already that $C_n \cong \mathbb{Z}/n\mathbb{Z}$ for each $n \in \mathbb{N}$, therefore we only need to determine the order of $\operatorname{Inn} C_p$ and $\operatorname{Aut} C_p$ in order to uniquely identify them up to an isomorphism. We'll prove that $\operatorname{Aut} C_p \cong (\mathbb{Z}/n\mathbb{Z})^{\times}$. Consider the map $\varphi\colon (\mathbb{Z}/n\mathbb{Z}) \to \operatorname{Aut} C_p$ defined as $\varphi(n) = \lambda g \in C_p\colon g^n$. We claim that φ is a group isomorphism. Denote $\lambda g \in C_p\colon g^n \in \operatorname{Aut} C_p$ by ψ_n.

   - Clearly:

     $$\varphi(xy) = \psi_{xy} = \lambda g \in C_p\colon g^{xy} = \lambda g \in C_p\colon g^x g^y = (\lambda g \in C_p\colon g^x) \cdot_{\operatorname{Aut}} (\lambda g \in C_p\colon g^y) = \psi_x \psi_y = \varphi(x)\varphi(y)$$

     Hence φ is an isomorphism.

   - φ is clearly one-to-one:

     $$\psi_k = \psi_n \iff \forall g \in C_p\colon g^n = g^k \iff \forall z \in \mathbb{Z}\colon z^n \bmod p = z^k \bmod p \iff n \equiv k$$

     Yet $n, k \in (\mathbb{Z}/n\mathbb{Z})^{\times}$ therefore $n = k$.

   - φ also onto. Consider some $\psi \in \operatorname{Aut} C_p$, and we'll prove ψ has the form of ψ_n for some $n \in [n]$. Denote the generator of C_p with g. Each $t \in C_p$ has the form $t = g^{k_t}$ for some $k_t \in \mathbb{N}$, hence (since t is an homomorphism):

     $$\psi(t) = \psi(g^{k_t}) = \psi(g)^{k_t} = g^{k_t \cdot k_{\psi(g)}} = t^{k_{\psi(g)}} = \psi_{k_{\psi(g)}}$$

     Moreover, since ψ must have a trivial kernel, $k_{\psi(g)} \neq 0$ (otherwise $\psi(g) = g^{k_{\psi(g)}} = g^0 = e$ and $\psi(C_p)$ is generated by a neutral element, hence the trivial group). Therefore we can safely WOLOG assume that $k_{\psi(g)} \in (\mathbb{Z}/n\mathbb{Z})^{\times}$ as intended.

Since p is prime, we know that $(\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$. Finally:

$$\mathbf{Aut}\, \boldsymbol{C_p} \cong \mathbb{Z}_p^\times \cong \mathbb{Z}/(p-1)\mathbb{Z} \cong \boldsymbol{C_{p-1}}$$

As intended.

As for inner automorphisms: $\operatorname{Aut} C_p \ni \psi_n(g) = t^n$ has the form of $t \mapsto a^{-1}ta$ if and only if $a = e$ and $n = 1$, as cyclic groups are Abelian and therefore $t \mapsto a^{-1}ta = t$, the identity map. Hence $\mathbf{Inn}\, \boldsymbol{C_p} \cong \boldsymbol{\{e\}}$.

4. Consider H, K be groups, and $\varphi\colon K \to \operatorname{Aut} H$ homomorphism. The semi-definite product of H and K by φ is:

$$H \rtimes_\varphi K \coloneqq \{(h,k)\colon h \in H, k \in K\} \qquad (h_1,k_1)\cdot(h_2,k_2) \coloneqq (h_1 \cdot (\varphi(k_1))(h_2), k_1k_2)$$

(a) We'll prove it's indeed a valid group.

Proof. Let φ be such homomorphism, then:

- **Multiplication is associative:**

$$\begin{aligned}
&(h_1,k_1)\cdot((h_2,k_2)\cdot(h_3,k_3)) \\
&= (h_1,k_1)\cdot(h_2\cdot(\varphi(k_2))(h_3), k_2k_3) \\
&= (h_1\cdot(\varphi(k_1))\big(h_2\cdot(\varphi(k_2))(h_3)\big),\ k_1\,(k_2k_3)) \qquad \varphi \textit{ homomorphism} \\
&= (h_1\cdot(\varphi(k_1))(h_2)\cdot(\varphi(k_1))(\varphi(k_2)(h_3),\ (k_1k_2)\,k_3) \\
&= (h_1\cdot(\varphi(k_1))(h_2), k_1k_2)\cdot(h_3,k_3) \\
&= ((h_1,k_1)\cdot(h_2,k_2))\cdot(h_3,k_3)
\end{aligned}$$

- **Unitary element exists:**

$$\underbrace{(e_H,e_K)}_{e_{H\rtimes K}}\cdot(h,k) = \big(e_H\cdot\underbrace{(\varphi(e_H))}_{e_{\operatorname{Aut} H}}(h), e_K\cdot k\big) = (e_H\cdot h,\ e_K\cdot k) = (h,k)$$

- **Inverse exists:** for all $(h,k) \in H \rtimes_\varphi K$, since $\varphi(k^{-1})$ is an automorphism and in particular bijective, there exists an element denoted $\tilde{h}^{-1}$ s.t. $(\varphi(k^{-1}))(\tilde{h}^{-1}) = h^{-1}$. Therefore:

$$(h,k)\cdot(\tilde{h}^{-1},k^{-1}) = \big(h.(\varphi(k^{-1}))(\tilde{h}^{-1}), k\cdot k^{-1}\big) = (h\cdot h^{-1}, k\cdot k^{-1}) = (e_H,e_K) = e_{H\rtimes_\varphi K}$$

■

(b) We'll proceed by proving that $H \cong H\times\{e\} \lhd H\rtimes_\varphi K$.

Proof. Denote $G = H\rtimes_\varphi K$. Let $g \in G$, we'll prove that $(H\times\{e\})^{g^{-1}} = H$ (equations are up to a natural isomorphism). Let $\hbar \in H$, and denote $g = (h,k)$.

$$\begin{aligned}
&g\cdot(\hbar,e_K)\cdot g^{-1} \\
&= (h,k)\cdot(\hbar,e_K)\cdot(\tilde{h}^{-1},k^{-1}) \\
&= \big(h\cdot(\varphi(k))(\hbar\cdot(\varphi(e)(\tilde{h}^{-1}))),\ k\cdot e_K\cdot k^{-1}\big) \\
&= \big(h\cdot(\varphi(k))(\hbar)\cdot(\varphi(k))((e_{\operatorname{Aut} H})\tilde{h}^{-1}), e_K\big) \\
&= \big(h\cdot(\varphi(k))(\hbar)\cdot h^{-1}, e_K\big) \subset H\times\{e_K\}
\end{aligned}$$

Now clearly $H^{g^{-1}} \subset H$ for all $g \in G$. Therefore $\forall g\in G\colon H^g \subset H$ which implies that $\forall g \in G\colon (H^g)^{g^{-1}} \subset H^{g^{-1}}$, hence $H \subset H^{g^{-1}}$ too. Finally $\forall g\in G\colon H^{g^{-1}} = H$.

Clearly, a group is normal iff $\forall g\in G\colon H^g = H$ iff $\forall g\in G\colon H^{g^{-1}} = H$, hence $H\times\{e\}$ is normal. ■

(c) Consider φ the trivial homomorphism. Then $H\rtimes_\varphi K \cong H\times K$.

Proof. This is in fact a strict equality, since both groups are equivalent under the forgetful functor $\mathbf{Grp} \to \mathbf{Set}$ and their operations behaves exactly the same:

$$(h_1,k_1)\cdot_{\rtimes_\varphi}(h_2,k_2) = (h_1\cdot\underbrace{(\varphi(k_1))}_{e_{\operatorname{Aut} H}}(h_2), k_1k_2) = (h_1h_2,k_1k_2) = (h_1,k_1)\cdot_\times(h_2,k_2)$$

■

5. Let G be a group, and let $H, K \leq G$ subgroups s.t. $H \lhd G, H \cap K = \{e_G\}, HK = G$. We'll prove that $G \cong H \rtimes_\varphi K$ for some $\varphi\colon K \to \operatorname{Aut} H$.

   Proof. content ■